



NORTHERN AREAS

Challenges for reconstruction in remote
areas

A photograph of a mountain landscape. In the background, a jagged mountain peak is covered in snow and partially obscured by white clouds. The sky is blue with scattered white clouds. The foreground and middle ground show a steep, brownish-yellow hillside with sparse vegetation and a dirt path winding through it. The overall scene is rugged and high-altitude.

WHAT DO WE MEAN BY NORTHERN?

- ▶ “Northern” as a proxy for:
 - ▶ Remote
 - ▶ High altitude – often low vegetation
 - ▶ Linguistic or Ethnic minority
 - ▶ High cost
 - ▶ Low service

HIGH DAMAGE LEVELS

- Unsafe pre-earthquake housing stock pre-Earthquake
- EQ-resilience not a feature of many vernacular styles, despite high skill levels in stone work & carpentry!
- Dry stone / mud mortar
- Low availability of other materials
- Life & construction in remote areas hugely expensive



FAST REBUILDING

- Culture of self-sufficiency and recovery – by necessity
- Low expectations of government support
- Low temperatures
- No timely information or culturally appropriate designs
- Absence of alternatives – materials, models, advice
- Rebuilding with salvaged materials



BUILDING BACK BETTER? BUILDING BACK WORSE?



Evidence of some personal learning - but hugely variable,
and
Few structures built to code



BUILDING BACK BETTER? BUILD BACK WORSE?

- The further from the road, the greater the vulnerability
- In some cases, repair of structurally-compromised buildings
- In others, entire villages of toppled buildings put back up again, like puzzle pieces
- Some communities have built back worse - e.g. without windows
- Some examples of improved techniques where DUDBC Engineers have engaged
- Rebuilding on landslide-affected land

THE CHALLENGE OF REMOTENESS

- No road access - porters, mules, helicopters - high costs
- Damaged trails
- Local materials - wood, stone, slate, sand, water - hugely variable availability and cost
- Vast differences within the same VDCs – complicates value chain or market assessments
- Labour brought in from outside, and associated costs/complications
- No access to finance, limited access to services, high interest rates
- High expenditure to reach DHQs to withdraw money
- Lower levels of formal education + literacy – intersection of physical and conceptual marginalization
- Language barriers affect interaction w/ state + access to info – even from our partners

CULTURAL DIVERSITY VS. THE 17 MODELS



- 17 models not feasible, suitable or desirable – immediately recognizable as ‘foreign’ or ‘Kathmandu’ houses
- Accessing housing grant means changing culture
- Lack of recognition, alienation, fear of erasure
- Homes reflect livelihoods, marriage practices & family structures
- Two-room boxes / Identikit NGO support
- The need: SAFE VERNACULAR



DUDBC ENGINEERS

- From outside the VDC – vulnerable to pressure, unlikely to speak the local language, comparatively young and inexperienced.
- Quality of training and advice highly variable, little formal mechanism for oversight or accountability (VDC Secretaries, themselves often in DHQ) – what about after LLR?
- Misleading advice - “Don’t use slate on roofs,” “just rebuild and we will approve it later.”
- Need to be proactively circulating throughout the VDC
- Retention remains a challenge